# Differences in Inspired & Expired Air

**Exam code:** 0610 & 0970

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## Differences in inspired & expired air

- Air that is inhaled, or breathed in, **differs in its gas composition** to air that is exhaled, or breathed out; this is due to the process of **gas exchange** that takes place in the alveoli
  - Inhaled air can also be referred to as inspired air
  - Exhaled air is also known as expired air
- Inhaled air is drawn from the surrounding atmosphere, and so its gas composition matches atmospheric levels
- During gas exchange in the alveoli **oxygen enters the blood** from the alveoli, and **carbon dioxide** and **water vapour leave the blood** and enter the alveoli
- This gas exchange process means that the gas composition of exhaled air differs to that of the air that was previously inhaled
  - Inhaled air contains around **21 % oxygen** and exhaled air contains around **16 % oxygen**
  - Inhaled air contains around **0.04 % carbon dioxide** and exhaled air contains around **4 % carbon dioxide**
  - Inhaled air contains less water vapour than exhaled air

**Composition of inhaled and exhaled air table**

| Gas | Inspired air / % | Expired air / % |
|---|---|---|
| Oxygen | 21 | 16 |
| Carbon dioxide | 0.04 | 4 |
| Water vapour | 0.2 - 4 | 1 - 5 |
| Nitrogen | 78 | 78 |

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